

GRADE 12 MATHEMATICS

EUEE MASTER STUDY GUIDE

Ethiopian University Entrance Exam
Complete Exam Preparation · All 5 Units

Sequences & Series · Calculus · Statistics
Linear Programming · Business Mathematics

Simplified · Exam-Focused · High-Score Strategies

This guide covers every topic in your Grade 12 Mathematics textbook, simplified and organised so you can study smarter — not harder. All formulas are written with proper mathematical symbols ($\times$, $\div$, $\sqrt{}$, $\sum$, $\int$, σ ...) so they look exactly as they would in an exam. Here is what each coloured box means:

EXAM FOCUS	Must-know content — these topics appear in almost every EUEE paper.
COMMON MISTAKES	The errors that cost students marks every single year.
PRO TIP	Memory tricks and smart strategies to work faster and more accurately.
QUICK REVISION	Flash-card style summaries — perfect for the night before the exam.
KEY FORMULAS	Every formula you must memorise, written clearly with proper symbols.
PRACTICE QUESTIONS	EUEE-style questions to test your understanding immediately.

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- 1.2 Arithmetic and Geometric Sequences
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UNIT 1

SEQUENCES AND SERIES

1.1 Sequences

A sequence is simply an ordered list of numbers that follows a clear rule. Each number in the list is called a **term**. Knowing the rule lets you find any term without writing out the whole list.

Definition — Sequence

A function whose input is the set of positive integers. The terms are written $a_1, a_2, a_3, \dots, a_n, \dots$. The term a_n is called the **general term** or the **n-th term**.

Two types of sequences:

- **Finite sequence** — has a definite last term. Example: 2, 4, 6, 8, 10
- **Infinite sequence** — goes on forever (no last term). Example: 1, 3, 5, 7, 9, ...

Recursion formula

A recursion formula defines each new term using the one before it. For example, if $a_1 = 3$ and each next term adds 4, you get 3, 7, 11, 15, ...

Example 1.1

Given $a_n = 3n - 1$, find the first four terms.

Substitute $n = 1, 2, 3, 4$ one at a time:

$$a_1 = 3(1) - 1 = \mathbf{2}, a_2 = 3(2) - 1 = \mathbf{5}, a_3 = 3(3) - 1 = \mathbf{8}, a_4 = 3(4) - 1 = \mathbf{11}$$

The sequence is: 2, 5, 8, 11, ...

EXAM FOCUS

- Find the n-th term when you are given the first few terms — very common in EUEE.
- Also practice listing terms when you are given the general formula a_n .
- Recursion formulas appear often — make sure you can use the previous term.

1.2 Arithmetic and Geometric Sequences

Arithmetic Sequence (AP)

Definition — Arithmetic Sequence (AP)

Each term is found by **adding the same fixed number** to the previous term. That fixed number is called the **common difference d**. Example: 3, 7, 11, 15, ... has $d = 4$.

KEY FORMULAS

$$a_n = a_1 + (n - 1) \times d$$

General term — tells you the value of any term directly.

$$d = a_2 - a_1$$

Common difference — subtract any term from the one after it.

$$S_n = n/2 \times (a_1 + a_n)$$

Sum of the first n terms — use when you know the first and last term.

$$S_n = n/2 \times [2a_1 + (n - 1) \times d]$$

Sum of the first n terms — use when you only know a_1 and d .

$$\text{Arithmetic mean of } a \text{ and } b = (a + b) \div 2$$

The middle value that sits exactly between a and b .

Example 1.2

An AP has $a_1 = 5$ and $d = 3$. Find a_{10} and S_{10} .

$$a_{10} = 5 + (10 - 1) \times 3 = 5 + 27 = \mathbf{32}$$

$$S_{10} = 10/2 \times (5 + 32) = 5 \times 37 = \mathbf{185}$$

Example 1.3

How many terms are in the AP: 7, 11, 15, ..., 91?

$$d = 4. \text{ Use } a_n = a_1 + (n - 1) \times d: 91 = 7 + (n - 1) \times 4 \rightarrow 84 = (n - 1) \times 4 \rightarrow n - 1 = 21 \rightarrow n = \mathbf{22}$$

Geometric Sequence (GP)

Definition — Geometric Sequence (GP)

Each term is found by **multiplying the previous term by the same fixed number**. That fixed number is called the **common ratio r** . Example: 2, 6, 18, 54, ... has $r = 3$.

KEY FORMULAS

$$a_n = a_1 \times r^{n-1}$$

General term — the exponent is always one less than the term number.

$$r = a_2 \div a_1$$

Common ratio — divide any term by the one before it.

$$S_n = a_1 \times (r^n - 1) \div (r - 1)$$

Sum of n terms — use this when $r \neq 1$.

$$\text{Geometric mean of } a \text{ and } b = \sqrt{a \times b}$$

The middle value that sits between a and b in a GP.

Example 1.4

A GP has $a_1 = 3$ and $r = 2$. Find a_6 and S_6 .

$$a_6 = 3 \times 2^{6-1} = 3 \times 32 = \mathbf{96}$$

$$S_6 = 3 \times (2^6 - 1) \div (2 - 1) = 3 \times 63 \div 1 = \mathbf{189}$$

COMMON MISTAKES

- Using $a_n = a_1 \times r^n$ instead of $a_1 \times r^{n-1}$ — the exponent starts at zero, not one.
- Confusing arithmetic mean $(a + b) \div 2$ with geometric mean $\sqrt{a \times b}$.
- Forgetting to check whether $r > 1$ or $r < 1$ before summing a GP.

PRO TIP

- AP: look for a number being **added** each time. GP: look for a number being **multiplied** each time.
- If $d > 0$ the AP goes up. If $d < 0$ it goes down.
- If $|r| > 1$ the GP grows without limit. If $|r| < 1$ it shrinks toward zero.

1.3 Sigma Notation and Partial Sums

Sigma notation is a shorthand way to write a long sum using the Greek capital letter Σ (pronounced "sigma"). It says: add up every term from the lower limit to the upper limit.

Definition — Sigma Notation

$S_n = \sum_{k=1}^n a_k = a_1 + a_2 + a_3 + \dots + a_n$ where k is the index, 1 is the lower limit, and n is the upper limit. S_n is called the **n-th partial sum**.

KEY FORMULAS

$$\sum_{k=1}^n k = n(n+1) \div 2$$

Sum of the first n whole numbers. Example: $1+2+3+\dots+100 = 5\,050$.

$$\sum_{k=1}^n k^2 = n(n+1)(2n+1) \div 6$$

Sum of the first n perfect squares.

$$\sum_{k=1}^n k^3 = [n(n+1) \div 2]^2$$

Sum of the first n perfect cubes.

$$\sum (c \times a_k) = c \times \sum a_k$$

You can factor out a constant from a sigma.

Example 1.5

Evaluate $\sum_{k=1}^5 (2k + 1)$.

Break it up: $\sum (2k + 1) = 2 \times \sum k + \sum 1$

$\sum k$ from 1 to 5 = $5 \times 6 \div 2 = 15$. $\sum 1$ from 1 to 5 = 5.

Answer: $2 \times 15 + 5 = 30 + 5 = \mathbf{35}$

EXAM FOCUS

- Practice writing a series in sigma notation AND expanding sigma back to a sum — both directions are tested.
- Know $\sum k$, $\sum k^2$, and $\sum k^3$ by heart — they save a lot of time.
- Partial sum of an AP: $S_n = n/2 \times [2a_1 + (n-1) \times d]$
- Partial sum of a GP: $S_n = a_1 \times (r^n - 1) \div (r - 1)$

1.4 Infinite Series

Definition — Infinite Series

An infinite series is the sum of infinitely many terms: $S = a_1 + a_2 + a_3 + \dots$. If the partial sums get closer and closer to a fixed value L as you add more terms, the series **converges** to L . Otherwise it **diverges** (no finite sum exists).

KEY FORMULAS

$$S_{\infty} = a_1 \div (1 - r)$$

Sum of an infinite GP — only valid when $|r| < 1$.

If $|r| \geq 1$, the infinite GP diverges — there is no finite sum.

Example 1.6

Find the sum of the infinite series: $8 + 4 + 2 + 1 + \dots$

$a_1 = 8$, $r = 4 \div 8 = \frac{1}{2}$. Since $|\frac{1}{2}| < 1$, the series converges.

$$S_{\infty} = 8 \div (1 - \frac{1}{2}) = 8 \div \frac{1}{2} = \mathbf{16}$$

Example 1.7 — Repeating decimal

Write $0.333\dots$ as a fraction.

$$0.333\dots = \frac{3}{10} + \frac{3}{100} + \frac{3}{1\,000} + \dots$$

$$a_1 = \frac{3}{10}, r = \frac{1}{10}. S_{\infty} = (\frac{3}{10}) \div (1 - \frac{1}{10}) = (\frac{3}{10}) \div (\frac{9}{10}) = \mathbf{\frac{1}{3}}$$

PRO TIP

- Repeating decimals are always infinite GPs — a classic EUEE question type.
- Always find r first: $r = (\text{any term}) \div (\text{the term before it})$.
- Check $|r| < 1$ BEFORE using the infinite sum formula.

1.5 Applications in Daily Life

Sequences and series model many real situations. The key skill is deciding whether you have an AP or a GP, then choosing the right formula.

- **Population growth:** usually a GP (growing by a fixed percentage each year).
- **Salary increases:** an AP if a fixed amount is added each year.
- **Stacking rows:** an AP if each row has a fixed number more than the last.
- **Loans and investments:** usually a GP if a fixed percentage applies.

Example 1.8 (EUEE-style)

A farmer plants 50 trees in year 1 and increases by 15 each year. How many in year 8? What is the total over 8 years?

This is an AP with $a_1 = 50$ and $d = 15$.

$$a_8 = 50 + 7 \times 15 = 50 + 105 = \mathbf{155 \text{ trees}}$$

$$S_8 = \frac{8}{2} \times (50 + 155) = 4 \times 205 = \mathbf{820 \text{ trees total}}$$

QUICK REVISION

- **AP general term:** $a_n = a_1 + (n - 1) \times d$
- **AP sum:** $S_n = n/2 \times (a_1 + a_n)$ or $n/2 \times [2a_1 + (n - 1)d]$
- **GP general term:** $a_n = a_1 \times r^{n-1}$
- **GP sum:** $S_n = a_1 \times (r^n - 1) \div (r - 1)$
- **Infinite GP:** $S_\infty = a_1 \div (1 - r)$, only when $|r| < 1$
- **Σk** $= n(n+1)/2$ **Σk^2** $= n(n+1)(2n+1)/6$

PRACTICE QUESTIONS

1. An AP has $a_1 = 7$ and $d = 5$. Find a_{12} and S_{12} .
2. A GP has $a_1 = 4$ and $r = 3$. Find the 5th term and S_5 .
3. Find the sum of the infinite GP: $6 + 2 + \frac{2}{3} + \dots$
4. Evaluate: $\Sigma(k=1 \text{ to } 10) (3k - 2)$
5. A school enrolls 200 students in year 1 and 15 more each year. In which year will it reach 500?

UNIT 2

INTRODUCTION TO CALCULUS

Calculus is the mathematics of change. **Differential calculus** (derivatives) measures how fast things change. **Integral calculus** (integration) measures the total accumulation of change. Together they describe motion, growth, area, and countless real-world phenomena.

2.1 Introduction to Derivatives

What is a Derivative?

Definition — Derivative

The derivative of a function f at a point x is the **instantaneous rate of change** of f at that point. Geometrically it equals the slope of the tangent line to the curve. Written as $f'(x)$ or dy/dx .

KEY FORMULAS

$$f'(x) = \lim(h \rightarrow 0) [f(x + h) - f(x)] \div h$$

Formal definition — the slope of the tangent line as h shrinks to zero.

$$\text{Power Rule: } d/dx [x^n] = n \times x^{n-1}$$

The most used rule. Bring the power down, then reduce the power by 1.

$$\text{Constant Rule: } d/dx [c] = 0$$

The derivative of any plain number is always zero.

$$\text{Constant Multiple: } d/dx [c \times f(x)] = c \times f'(x)$$

Constants can be factored out before differentiating.

$$\text{Sum / Difference: } d/dx [f \pm g] = f' \pm g'$$

Differentiate each term separately.

$$\text{Product Rule: } d/dx [f \times g] = f' \times g + f \times g'$$

For a product of two functions.

$$\text{Quotient Rule: } d/dx [f \div g] = (f' \times g - f \times g') \div g^2$$

For a fraction of two functions. ($g \neq 0$)

$$\text{Chain Rule: } d/dx [f(g(x))] = f'(g(x)) \times g'(x)$$

For a function inside another function — always multiply by the derivative of the inside.

Standard Derivatives — must know these by heart

$\sin x$	$\cos x$
$\cos x$	$-\sin x$

$\tan x$	$\sec^2 x$
e^x	e^x
$\ln x$	$1 \div x$
a^x	$a^x \times \ln a$
$\log_a x$	$1 \div (x \times \ln a)$

Example 2.1

Differentiate $f(x) = 4x^3 - 5x^2 + 3x - 7$

Apply the power rule to each term individually:

$$f'(x) = 4 \times 3x^2 - 5 \times 2x + 3 \times 1 - 0 = \mathbf{12x^2 - 10x + 3}$$

Example 2.2 — Product Rule

Find $f'(x)$ if $f(x) = (x^2 + 1)(3x - 2)$

Let $f = x^2 + 1$ and $g = 3x - 2$. Then $f' = 2x$ and $g' = 3$.

$$f'(x) = 2x(3x - 2) + (x^2 + 1)(3) = 6x^2 - 4x + 3x^2 + 3 = \mathbf{9x^2 - 4x + 3}$$

Example 2.3 — Chain Rule

Differentiate $f(x) = (3x^2 + 1)^5$

The "outside" is $(\dots)^5$ and the "inside" is $3x^2 + 1$.

$$f'(x) = 5(3x^2 + 1)^4 \times 6x = \mathbf{30x(3x^2 + 1)^4}$$

COMMON MISTAKES

- Chain Rule: $d/dx [(2x+1)^3] = 3(2x+1)^2 \times \mathbf{2} = 6(2x+1)^2$. Students often forget to multiply by the derivative of the inside (the 2).
- $d/dx [\sin x] = \mathbf{+\cos x}$, not $-\cos x$. Only $d/dx [\cos x]$ gives the minus sign.
- Power rule: $d/dx [x^5] = 5x^4$. The exponent drops from 5 to 4 — do not leave it as $5x^5$.

2.2 Applications of Derivatives

Tangent and Normal Lines

The derivative $f'(a)$ gives the slope of the **tangent line** at $x = a$. The **normal line** is perpendicular to the tangent, so its slope is $-1 \div f'(a)$.

KEY FORMULAS

Tangent at point $(a, f(a))$: $y - f(a) = f'(a) \times (x - a)$

This is the point-slope form of the tangent line equation.

Normal at point $(a, f(a))$: $y - f(a) = [-1 \div f'(a)] \times (x - a)$

Use this when the question asks for the normal (perpendicular) line.

Example 2.4

Find the equation of the tangent to $y = x^2 - 3x$ at $x = 2$.

$$f(2) = 4 - 6 = -2. \quad f'(x) = 2x - 3, \quad f'(2) = 1.$$

Tangent: $y - (-2) = 1 \times (x - 2) \rightarrow y = x - 4$

Increasing, Decreasing and Critical Points

Definition — Critical Number

A value c is a critical number if $f'(c) = 0$ or $f'(c)$ does not exist. Critical numbers are the candidates for local maximum and local minimum values.

KEY FORMULAS

f is increasing on any interval where $f'(x) > 0$

Positive derivative means the curve is going uphill.

f is decreasing on any interval where $f'(x) < 0$

Negative derivative means the curve is going downhill.

First Derivative Test: if f' changes $+$ to $- \rightarrow$ local MAX

First Derivative Test: if f' changes $-$ to $+$ $\rightarrow$ local MIN

Second Derivative Test: $f'(c) = 0$ and $f''(c) < 0 \rightarrow$ local MAX

Second Derivative Test: $f'(c) = 0$ and $f''(c) > 0 \rightarrow$ local MIN

Example 2.5

Find the local max and min of $f(x) = x^3 - 3x^2 - 9x + 5$.

Step 1 — Differentiate: $f'(x) = 3x^2 - 6x - 9 = 3(x - 3)(x + 1)$

Step 2 — Critical numbers: $x = 3$ and $x = -1$

Step 3 — Second derivative: $f''(x) = 6x - 6$

$f''(-1) = -12 < 0 \rightarrow$ local **maximum** at $x = -1$. $f(-1) = -1 - 3 + 9 + 5 = \mathbf{10}$

$f''(3) = 12 > 0 \rightarrow$ local **minimum** at $x = 3$. $f(3) = 27 - 27 - 27 + 5 = \mathbf{-22}$

EXAM FOCUS

- Tangent and normal line questions appear in nearly every EUEE exam — practise this template.
- To find absolute max/min on a closed interval $[a, b]$: evaluate f at every critical number AND at both endpoints. Largest = max, smallest = min.
- Optimisation word problems: write f , set $f' = 0$, find critical numbers, confirm max or min.

2.3 Introduction to Integration

Definition — Indefinite Integral

If $F'(x) = f(x)$, then F is called an **anti-derivative** of f . The indefinite integral is written $\int f(x) dx = F(x) + C$, where C is the **constant of integration** and must always be included.

KEY FORMULAS

Power Rule: $\int x^n dx = x^{n+1} \div (n+1) + C$

Increase the power by 1, then divide by the new power. Only works when $n \neq -1$.

$\int c dx = c \times x + C$

Integral of a constant.

$\int e^x dx = e^x + C$

$\int (1 \div x) dx = \ln|x| + C$

This is the special case where the power rule doesn't apply ($n = -1$).

$\int \sin x dx = -\cos x + C$

$\int \cos x dx = \sin x + C$

Definite Integral: $\int(a \text{ to } b) f(x) dx = F(b) - F(a)$

Fundamental Theorem of Calculus — substitute the limits and subtract.

Example 2.6

Find $\int (5x^3 - 4x + 7) dx$

Integrate each term separately using the power rule:

$= 5x^4/4 - 4x^2/2 + 7x + C = \mathbf{(5/4)x^4 - 2x^2 + 7x + C}$

Example 2.7 — Definite Integral

Evaluate $\int(1 \text{ to } 3) (2x + 1) dx$

First, find the anti-derivative: $F(x) = x^2 + x$

Then substitute limits: $F(3) - F(1) = (9 + 3) - (1 + 1) = 12 - 2 = \mathbf{10}$

Integration by Substitution

When the integrand contains a function inside another function, let u = the inside function, find du , substitute everything into u , integrate, then substitute back.

Example 2.8

Find $\int 2x \times (x^2 + 1)^4 dx$

Let $u = x^2 + 1$. Then $du = 2x dx$. The integral becomes:

$\int u^4 du = u^5 \div 5 + C = \mathbf{(x^2 + 1)^5 \div 5 + C}$

COMMON MISTAKES

- Forgetting $+ C$ in indefinite integrals loses marks every time — make it a habit.
- Definite integrals: it is $F(b) - F(a)$, not $F(a) - F(b)$. Upper limit first.
- $\int (1/x) dx$ is **not** $x^0 \div 0$ — that is undefined. The answer is $\ln|x| + C$.

PRO TIP

- Always verify your integral by differentiating the answer — you should get back $f(x)$.
- Do NOT write $+ C$ for definite integrals — the constant cancels when you subtract.

QUICK REVISION

- **Power Rule (diff):** $d/dx [x^n] = n \times x^{n-1}$
- **Chain Rule:** $d/dx [f(g(x))] = f'(g(x)) \times g'(x)$
- **Power Rule (int):** $\int x^n dx = x^{n+1} \div (n+1) + C$
- **Definite:** $\int(a \text{ to } b) f(x) dx = F(b) - F(a)$
- **Critical numbers:** set $f'(x) = 0$, then use f'' to classify max or min

PRACTICE QUESTIONS

1. Differentiate: $f(x) = 3x^4 - 7x^2 + 5x - 2$
2. Find the tangent line equation for $y = x^2 + 2x$ at $x = 1$.
3. Find all local max and min of $f(x) = 2x^3 - 9x^2 + 12x - 4$.
4. Evaluate: $\int(0 \text{ to } 2) (3x^2 - 2x + 1) dx$
5. Find: $\int (4x + 3)^5 dx$ using substitution.

UNIT 3

STATISTICS

Statistics helps us make sense of data. In Grade 12 we focus on **measures of dispersion** — how spread out the data values are — and on **sampling techniques** for studying large populations.

3.1 Measures of Absolute Dispersion

Absolute measures describe the spread of data in the same units as the data itself. They tell us whether data values are clustered together or widely scattered.

Range and Interquartile Range

KEY FORMULAS

Range = Maximum value – Minimum value

The simplest measure of spread — but affected a lot by extreme values.

IQR (Interquartile Range) = $Q_3 - Q_1$

The spread of the middle 50% of the data. Q_1 = lower quartile, Q_3 = upper quartile.

Quartile Deviation (QD) = $(Q_3 - Q_1) \div 2 = \text{IQR} \div 2$

Half the interquartile range — a stable measure not affected by extreme values.

Finding quartiles — step by step

1. Sort the data in ascending order.
2. Q_2 = median (middle value of the whole data set).
3. Q_1 = median of the lower half of the data.
4. Q_3 = median of the upper half of the data.

Example 3.1

Data: 4, 7, 10, 13, 16, 20, 25, 30, 35

$Q_1 = 10$, $Q_2 = 20$, $Q_3 = 30$

$\text{IQR} = 30 - 10 = \mathbf{20}$. $\text{QD} = 20 \div 2 = \mathbf{10}$

Mean Deviation

Definition — Mean Deviation (MD)

The average of the absolute distances each value is away from the mean. "Absolute" means we ignore the sign — we only care about how far, not which direction.

KEY FORMULAS

$$\text{MD} = (1 \div n) \times \sum |x_i - \bar{x}|$$

For raw data: find the mean $\bar{x}$, subtract from each value, take |absolute value|, average them.

$$\text{MD} = (1 \div N) \times \sum f_i \times |m_i - \bar{x}|$$

For grouped data: f_i = frequency, m_i = class midpoint, N = total frequency.

Example 3.2

Data: 6, 8, 10, 12, 14. Mean $\bar{x} = 10$

Deviations from mean: $|6-10| + |8-10| + |10-10| + |12-10| + |14-10|$

$$= 4 + 2 + 0 + 2 + 4 = 12$$

$$\text{MD} = 12 \div 5 = \mathbf{2.4}$$

Variance and Standard Deviation

KEY FORMULAS

$$\text{Variance } (\sigma^2) = (1 \div n) \times \sum (x_i - \bar{x})^2$$

Square each deviation from the mean, then find the average of those squares.

$$\text{Shortcut for variance: } \sigma^2 = (1/n \times \sum x_i^2) - \bar{x}^2$$

Often faster than computing each $(x_i - \bar{x})^2$ individually.

$$\text{Standard Deviation } (\sigma) = \sqrt{\text{Variance}} = \sqrt{\sigma^2}$$

Square root of variance — brings the unit back to the original unit of data.

$$\text{For grouped data: } \sigma^2 = (1 \div N) \times \sum f_i \times (m_i - \bar{x})^2$$

Example 3.3

Data: 2, 4, 4, 4, 5, 5, 7, 9 ($n = 8$, mean = 5)

Squared deviations: $(2-5)^2 + (4-5)^2 \times 3 + (5-5)^2 \times 2 + (7-5)^2 + (9-5)^2$

$$= 9 + 1 + 1 + 1 + 0 + 0 + 4 + 16 = 32$$

$$\sigma^2 = 32 \div 8 = 4. \sigma = \sqrt{4} = \mathbf{2}$$

EXAM FOCUS

- Know ALL four measures: Range, IQR, Mean Deviation, Standard Deviation.
- For grouped data, always use class midpoints — not class boundaries.
- Standard deviation = $\sqrt{\text{variance}}$. Variance = σ^2 . Do not mix them up.
- Both raw data and grouped data calculations appear in the EUEE.

3.2 Relative Measures of Dispersion

Relative measures let you compare the spread of two data sets that have different units or different means. They are expressed as ratios or percentages, so they have no units.

KEY FORMULAS

Coefficient of Range = $(\text{Max} - \text{Min}) \div (\text{Max} + \text{Min})$

Coefficient of QD = $(Q_3 - Q_1) \div (Q_3 + Q_1)$

Coefficient of MD = $\text{MD} \div \bar{x}$

Coefficient of Variation (CV) = $(\sigma \div \bar{x}) \times 100\%$

The most widely used. Lower CV means the data is more consistent.

Example 3.4

Class A: mean = 60, $\sigma = 8$. Class B: mean = 80, $\sigma = 12$.

$\text{CV(A)} = (8 \div 60) \times 100\% = 13.3\%$

$\text{CV(B)} = (12 \div 80) \times 100\% = 15.0\%$

Class A has the lower CV, so Class A is **more consistent**.

Skewness

Definition — Skewness

Skewness describes the shape of a distribution — whether it is symmetric or lopsided. A symmetric bell curve has mean = median = mode.

- **Right skew (positive):** mean > median > mode. Tail extends to the right.
- **Left skew (negative):** mean < median < mode. Tail extends to the left.
- **Symmetric:** mean = median = mode.

KEY FORMULAS

Pearson's Coefficient of Skewness = $3 \times (\text{Mean} - \text{Median}) \div \sigma$

Positive value → right-skewed. Negative → left-skewed. Zero → symmetric.

3.3 Frequency Curves

The **cumulative frequency curve** (also called an Ogive) is a smooth S-shaped graph used to read off quartiles, medians and percentiles directly from a chart.

- **Less-than Ogive:** cumulative frequency plotted against **upper** class boundaries. Slopes upward.
- **More-than Ogive:** cumulative frequency plotted against **lower** class boundaries. Slopes downward.
- **Reading the median:** draw a horizontal line at cumulative frequency = $N \div 2$. Where it hits the curve, read down to the x-axis.

3.4 Sampling Techniques

Definition — Population vs Sample

A **population** is every member of the group being studied. A **sample** is a smaller group drawn from the population for practical study. Good sampling ensures the sample represents the population fairly.

Random (Probability) Methods:

- **Simple Random:** every member has an equal chance of selection (lottery / random number table).
- **Systematic:** select every k-th member. $k = \text{Population size} \div \text{Sample size}$.
- **Stratified:** divide into groups (strata) by characteristics, then sample from each group proportionally.
- **Cluster:** divide into natural clusters (e.g. villages), then randomly select entire clusters.
- **Multi-stage:** combines several sampling methods in stages.

Non-Random Methods:

- **Convenience:** use whoever is easiest to reach.
- **Judgement (Purposive):** researcher hand-picks members based on expertise.

EXAM FOCUS

- Know every sampling method and when each is most appropriate — this is a guaranteed EUEE topic.
- Stratified is proportional: if 40% of the population is female, 40% of the sample must be female.
- Systematic: $N=500$, $n=50 \rightarrow$ select every $500 \div 50 = 10$ th person.

QUICK REVISION

- **Range** = $\text{Max} - \text{Min}$ **IQR** = $Q_3 - Q_1$ **QD** = $\text{IQR} \div 2$
- **MD** = $\sum |x_i - \bar{x}| \div n$ **σ^2** = $\sum (x_i - \bar{x})^2 \div n$ **σ** = $\sqrt{\sigma^2}$
- **CV** = $(\sigma \div \bar{x}) \times 100\%$ — lower CV means more consistent
- **Skewness:** right: $\text{mean} > \text{median} > \text{mode}$ left: $\text{mean} < \text{median} < \text{mode}$

PRACTICE QUESTIONS

1. Data: 3, 5, 7, 9, 11, 13, 15. Find Range, IQR and Quartile Deviation.
2. Find the standard deviation of: 10, 20, 30, 40, 50.
3. Factory A: mean = 50 mm, $\sigma = 2$ mm. Factory B: mean = 40 mm, $\sigma = 1.6$ mm. Which is more consistent?
4. A school has 400 students: 240 boys and 160 girls. A stratified sample of 50 is needed. How many boys?
5. Explain one advantage of stratified sampling over simple random sampling.

UNIT 4

INTRODUCTION TO LINEAR PROGRAMMING

Linear Programming (LP) finds the best possible outcome — maximum profit or minimum cost — from a set of real-world limitations expressed as linear inequalities. It is used in business, agriculture, engineering, and logistics.

4.1 Graphical Solutions of Linear Inequalities

Definition — Feasible Region

The feasible region is the set of all points (x, y) that satisfy **every** constraint at once. On a graph, it is the region where all the shaded areas overlap.

Steps to graph a system of inequalities:

1. Replace each inequality with an equation and draw the boundary line.
 2. Test the origin $(0, 0)$ — if it satisfies the inequality, shade that side.
 3. Repeat for every inequality.
 4. The feasible region is where ALL shaded areas overlap.
- **Solid line ($\leq$ or $\geq$):** points on the boundary line are included.
 - **Dashed line ($<$ or $>$):** points on the boundary line are excluded.

4.2 Maximum and Minimum Values

Definition — Objective Function

The linear function to be maximised or minimised, usually written $Z = ax + by$. It represents profit, cost, output, or whatever you are optimising.

The Corner Point Theorem

If the feasible region is **bounded** (enclosed), the maximum and minimum values of Z occur at one of the **corner points (vertices)** of the feasible region. Never anywhere else.

Step-by-step solution process:

1. Define your variables (e.g. x = units of Product A).
2. Write the objective function: $Z = ax + by$.
3. Write all constraints as inequalities.
4. Graph the feasible region.

5. Find all corner points by solving pairs of boundary line equations.
6. Substitute each corner point into Z .
7. Report the largest Z (maximum) or smallest Z (minimum).

Example 4.1 (Full EUEE-style problem)

Maximise $Z = 3x + 5y$ subject to: $x \geq 0$, $y \geq 0$, $x + y \leq 10$, $2x + y \leq 14$.

Draw the two boundary lines. The feasible region has four corners:

(0, 0)	0
(7, 0)	21
(4, 6)	$3 \times 4 + 5 \times 6 = 12 + 30 = 42$
(0, 10)	$3 \times 0 + 5 \times 10 = 50 \leftarrow \text{Maximum}$

The maximum value of $Z = 50$ occurs at the corner point **(0, 10)**.

Finding the intersection of two boundary lines:

Solve $x + y = 10$ and $2x + y = 14$ simultaneously. Subtract the first from the second: $x = 4$. Then $y = 10 - 4 = 6$. Point: (4, 6).

4.3 Real-Life Applications

The main challenge in application problems is translating English sentences into mathematical inequalities. Always identify what x and y represent first.

- **Production planning:** maximise profit given limits on labour, material and machine time.
- **Diet / nutrition:** minimise cost while meeting minimum nutritional needs.
- **Transport:** minimise shipping cost from warehouses to destinations.

COMMON MISTAKES

- Almost every LP problem requires $x \geq 0$ and $y \geq 0$. Do not forget these.
- Evaluate Z at every corner point — not just the ones that look biggest.
- Check your corner points satisfy all constraints before using them.

PRO TIP

- Always test (0, 0) first — it satisfies most constraints and is the easiest test point.
- To find a corner point: solve two boundary line equations simultaneously.

QUICK REVISION

- **LP process:** define $x, y \rightarrow$ write $Z = ax + by \rightarrow$ write constraints $\rightarrow$ graph $\rightarrow$ find corners $\rightarrow$ evaluate $Z \rightarrow$ report answer
- **Optimal value:** ALWAYS at a corner point of the feasible region.
- **Bounded region:** both a max and a min exist.
- **Unbounded region:** only one of max or min may exist — check carefully.

PRACTICE QUESTIONS

1. Graph the feasible region: $x \geq 0, y \geq 0, x + 2y \leq 8, 3x + y \leq 9$.
2. Maximise $Z = 4x + 3y$ subject to: $x \geq 0, y \geq 0, x + y \leq 5, 2x + y \leq 8$.
3. A factory makes chairs (x) and tables (y). Profit = $200x + 500y$. Constraints: $2x + 5y \leq 1000, x + y \leq 300$. Maximise profit.
4. Find the minimum of $Z = x + 2y$ on the region: $x \geq 0, y \geq 1, x + y \geq 4$.

UNIT 5

MATHEMATICAL APPLICATIONS IN BUSINESS

This unit applies mathematics to real-world money situations: percentages, ratios, interest, investments, loans, and taxation. These are highly practical topics and very popular with EUEE examiners.

5.1 Basic Business Concepts

Ratio, Proportion and Percentage

KEY FORMULAS

Ratio $a : b$ means $a \div b$

Equivalent ratios: if $a/b = c/d$ then $a \times d = b \times c$ (cross-multiply to check).

Direct proportion: $y = k \times x$

As x goes up, y goes up by the same factor.

Inverse proportion: $y = k \div x$

As x goes up, y goes down.

Percentage: $\text{Part} \div \text{Whole} \times 100$

Percentage increase = $(\text{New} - \text{Old}) \div \text{Old} \times 100\%$.

Markup amount = Selling Price – Cost Price

Markup % = $(\text{Markup} \div \text{Cost Price}) \times 100\%$.

Example 5.1

A shirt costs 300 Birr and is sold for 420 Birr. Find the markup %.

Markup = $420 - 300 = 120$ Birr. Markup% = $(120 \div 300) \times 100 = \mathbf{40\%}$

Depreciation

Definition — Depreciation

The decrease in value of an asset over time through use and wear. Appreciation is the opposite — an increase in value.

KEY FORMULAS

Straight-line depreciation per year = (Cost – Salvage Value) ÷ Useful Life

The asset loses the same fixed amount each year.

Book value at end of year n = Cost – $n \times$ Annual Depreciation

Declining balance: Value after n years = Cost $\times (1 - r)^n$

The asset loses the same percentage r of its current value each year.

Example 5.2

Machine costs 80 000 Birr. Salvage value 8 000 Birr. Useful life 8 years.

Annual depreciation = $(80\,000 - 8\,000) \div 8 = \mathbf{9\,000 \text{ Birr/year}}$

Book value at end of year 3 = $80\,000 - 3 \times 9\,000 = \mathbf{53\,000 \text{ Birr}}$

5.2 Time Value of Money

A Birr today is worth more than a Birr a year from now, because today's Birr can earn interest. This simple idea drives all financial mathematics.

Simple Interest

KEY FORMULAS

Interest: $I = P \times r \times t$

P = principal, r = annual interest rate (as a decimal), t = time in years.

Amount (Future Value): $A = P + I = P \times (1 + r \times t)$

The total you receive or owe at the end.

Example 5.3

5 000 Birr at 8% simple interest for 3 years.

$I = 5\,000 \times 0.08 \times 3 = 1\,200 \text{ Birr}$. $A = 5\,000 + 1\,200 = \mathbf{6\,200 \text{ Birr}}$

Compound Interest

With compound interest, the interest you earn is added to the principal, so next period you earn interest on a larger amount — interest on interest.

KEY FORMULAS

Compound Amount: $A = P \times (1 + r \div m)^{m \times t}$

P = principal, r = annual rate, m = compounding periods per year, t = years.

Compound Interest earned: $I = A - P$

Effective Annual Rate: $\text{EAR} = (1 + r \div m)^m - 1$

Converts any compounding frequency to an equivalent annual rate.

Present Value: $PV = A \times (1 + r \div m)^{-m \times t}$

How much you need to invest today to reach a target amount A in the future.

Annually	1
Semi-annually	2
Quarterly	4
Monthly	12
Daily	365

Example 5.4

10 000 Birr invested at 12% per year, compounded quarterly, for 5 years.

$$A = 10\,000 \times (1 + 0.12 \div 4)^{4 \times 5} = 10\,000 \times (1.03)^{20}$$

$$(1.03)^{20} \approx 1.8061. A \approx \mathbf{18\,061\,Birr}$$

5.3 Annuities

Definition — Annuity

A series of **equal payments** made at **regular intervals**. Ordinary annuity: payments at the END of each period. Annuity due: payments at the BEGINNING of each period.

KEY FORMULAS

Future Value of Ordinary Annuity: $FV = R \times [(1 + i)^n - 1] \div i$

R = regular payment, i = interest rate per period, n = total number of payments.

Present Value of Ordinary Annuity: $PV = R \times [1 - (1 + i)^{-n}] \div i$

How much all future payments are worth in today's money.

Annuity Due: multiply either formula by (1 + i)

Because payments start one period earlier.

Example 5.5

2 000 Birr saved at the end of each year for 5 years at 10% annual interest.

$$FV = 2\,000 \times [(1.1)^5 - 1] \div 0.10$$

$$= 2\,000 \times [1.6105 - 1] \div 0.10 = 2\,000 \times 6.105 = \mathbf{12\,210\,Birr}$$

Loan Repayment (Amortisation)

KEY FORMULAS

Monthly/Periodic payment: $R = PV \times i \div [1 - (1 + i)^{-n}]$

Use this to find how much you pay each period to clear a loan of amount PV.

Example 5.6

Loan: 50 000 Birr. Repaid monthly over 3 years. Annual rate: 12%.

$$i = 12\% \div 12 = 1\% = 0.01 \text{ per month. } n = 36 \text{ months.}$$

$$R = 50\,000 \times 0.01 \div [1 - (1.01)^{-36}]$$

$$= 500 \div [1 - 0.6989] = 500 \div 0.3011 \approx \mathbf{1\,661\,Birr/month}$$

5.4 Taxation

Definition — Tax

A compulsory payment made by individuals and businesses to the government, used to fund public services like education, health and roads.

Types of tax:

- **Income tax:** paid on personal earnings. Ethiopia uses a progressive system — higher income means higher rate.
- **VAT (Value Added Tax):** added at each stage of production. Ethiopia VAT rate = 15%.
- **Withholding tax:** deducted at source before you receive payment.
- **Profit tax:** paid by businesses on their net profits.

KEY FORMULAS

VAT amount = Taxable amount $\times$ 0.15

Ethiopia VAT rate is 15%.

Total price including VAT = Cost $\times$ 1.15

Net income = Gross income – Income tax – Other deductions

0 — 600	0% (exempt)
601 — 1 650	10%
1 651 — 3 200	15%
3 201 — 5 250	20%
5 251 — 7 800	25%
7 801 — 10 900	30%
Above 10 900	35%

Example 5.7

An employee earns 4 500 Birr per month. Calculate their income tax.

Apply the brackets progressively:

- 0 — 600: $600 \times 0\% = 0$ Birr
- 601 — 1 650: $1\,050 \times 10\% = 105$ Birr
- 1 651 — 3 200: $1\,550 \times 15\% = 232.50$ Birr
- 3 201 — 4 500: $1\,300 \times 20\% = 260$ Birr

Total tax = $0 + 105 + 232.50 + 260 = \mathbf{597.50}$ Birr

Net income = $4\,500 - 597.50 = \mathbf{3\,902.50}$ Birr

QUICK REVISION

- **Simple interest:** $I = P \times r \times t$ $A = P(1 + rt)$
- **Compound:** $A = P \times (1 + r/m)^{mt}$ $PV = A \times (1 + r/m)^{-mt}$
- **Annuity FV:** $R \times [(1+i)^n - 1] \div i$
- **Annuity PV:** $R \times [1 - (1+i)^{-n}] \div i$
- **VAT:** 15% in Ethiopia. $\text{Total} = \text{Cost} \times 1.15$

PRACTICE QUESTIONS

1. Find the compound interest on 20 000 Birr at 6% per year compounded semi-annually for 4 years.
2. A car worth 300 000 Birr depreciates at 20% per year (declining balance). Value after 3 years?
3. Monthly payment for a 100 000 Birr loan at 18% annual interest (monthly compounding) over 5 years?
4. Goods worth 45 000 Birr are sold. Calculate the VAT and the total price.
5. An employee earns 8 500 Birr/month. What is their income tax? What is their net income?

LAST-MINUTE REVISION GUIDE

You are almost there. Go through every table below. If anything feels unfamiliar, flip back to that unit and re-read the Key Formulas box. Then come back and read this page again.

UNIT 1 — SEQUENCES & SERIES

AP general term	$a_n = a_1 + (n - 1) \times d$
AP sum	$S_n = n/2 \times (a_1 + a_n)$ or $n/2 \times [2a_1 + (n-1)d]$
GP general term	$a_n = a_1 \times r^{n-1}$
GP sum	$S_n = a_1 \times (r^n - 1) \div (r - 1)$
Infinite GP	$S_\infty = a_1 \div (1 - r)$, only when $ r < 1$
$\sum k$	$n(n + 1) \div 2$
$\sum k^2$	$n(n + 1)(2n + 1) \div 6$
Arithmetic mean	$(a + b) \div 2$
Geometric mean	$\sqrt[n]{a \times b}$

UNIT 2 — CALCULUS

Power Rule (diff)	$d/dx [x^n] = n \times x^{n-1}$
Chain Rule	$d/dx [f(g(x))] = f'(g(x)) \times g'(x)$
Product Rule	$d/dx [f \times g] = f'g + fg'$
Quotient Rule	$d/dx [f \div g] = (f'g - fg') \div g^2$
$d/dx [\sin x]$	$\cos x$
$d/dx [\cos x]$	$-\sin x$
$d/dx [e^x]$	e^x
$d/dx [\ln x]$	$1 \div x$
$\int x^n dx$	$x^{n+1} \div (n+1) + C$
$\int (1/x) dx$	$\ln x + C$
$\int e^x dx$	$e^x + C$

Definite integral	$\int(a \text{ to } b) f(x) \, dx = F(b) - F(a)$
Critical numbers	Set $f'(x) = 0$, solve for x
2nd derivative test	$f''(c) < 0 \rightarrow \text{max } f''(c) > 0 \rightarrow \text{min}$

UNIT 3 — STATISTICS

Range	Maximum – Minimum
IQR	$Q_3 - Q_1$
QD	$(Q_3 - Q_1) \div 2$
Mean Deviation	$(1/n) \times \sum x_i - \bar{x} $
Variance σ^2	$(1/n) \times \sum (x_i - \bar{x})^2$
Std Deviation σ	$\sqrt{\text{variance}}$
Coeff. of Var.	$(\sigma \div \bar{x}) \times 100\%$
Pearson's SK	$3 \times (\text{Mean} - \text{Median}) \div \sigma$
Systematic k	Population size $\div$ Sample size

UNIT 4 — LINEAR PROGRAMMING

Define x and $y \rightarrow$ write $Z = ax + by \rightarrow$ write constraints $\rightarrow$ graph feasible region $\rightarrow$ find corner points $\rightarrow$ evaluate Z at each $\rightarrow$ state max or min.

Remember: the optimal value of Z ALWAYS occurs at a **corner point** of the feasible region.

UNIT 5 — BUSINESS MATHEMATICS

Simple interest	$I = P \times r \times t$ $A = P(1 + rt)$
Compound amount	$A = P \times (1 + r/m)^{mt}$
Present value	$PV = A \times (1 + r/m)^{-mt}$
Annuity FV	$R \times [(1+i)^n - 1] \div i$
Annuity PV	$R \times [1 - (1+i)^{-n}] \div i$
Loan payment R	$PV \times i \div [1 - (1+i)^{-n}]$
Straight-line depr.	(Cost – Salvage) $\div$ Useful life
Declining balance	$\text{Cost} \times (1 - r)^n$

VAT	15% in Ethiopia. Total = Cost × 1.15
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EUEE EXAM STRATEGY

Time Management

- **Skim first (5 minutes):** Read all questions quickly. Mark easy ones (E), medium (M) and hard (H).
- **Easy questions first:** Answer all E questions first — this builds confidence and locks in safe marks.
- **Pace yourself:** EUEE Maths: aim for roughly 2 minutes per question. If stuck for more than 3 minutes, move on and come back.
- **Show all working:** Even if your final answer is wrong, correct steps shown earn partial marks.
- **Check units:** Always confirm whether your answer should be in Birr, years, items, etc.
- **Final review:** Spend the last 10 minutes checking calculations — especially signs and substitutions.

Answering Techniques

- Multiple-choice: eliminate clearly wrong answers first. Then choose from what remains.
- Calculation questions: always write the formula first, then substitute — this earns partial marks even if the arithmetic slips.
- Graph questions (LP): draw neatly, label axes, shade feasible region, and list all corner points clearly.
- Proof / explanation: define terms clearly, then build the argument step by step.
- Statistics: show the mean calculation first, then the dispersion — examiners expect this order.
- Word problems (business maths): write down what is given, what is unknown, then choose the formula.

Common Exam Traps

- Wrong formula: AP uses d (common difference), GP uses r (common ratio). Check which one before you start.
- Calculus: forgetting $+ C$ in indefinite integrals costs marks every time — make it automatic.
- Statistics: standard deviation $= \sqrt{\text{variance}}$. Do not give variance when asked for σ .
- LP: evaluating Z at a point that is NOT a corner point of the feasible region — always use corners.

- Business: confusing simple interest and compound interest — look for the word "compounded" in the question.
- Units mismatch: rate given per month, time given in years — convert before you calculate.
- Tangent line: use $f'(a)$ for slope, not $f(a)$. Never substitute into the original function.

Most Important Topics to Revise

Sequences	General term (AP & GP), sum formulas, infinite GP convergence.
Derivatives	Power rule, chain rule, product rule, standard derivatives table.
Optimisation	Critical numbers, 1st/2nd derivative tests, tangent line equation.
Integration	Standard integrals, definite integral evaluation, area under curve.
Statistics	σ , CV, skewness, all four sampling methods.
LP	Setting up constraints, graphing, corner point theorem.
Business maths	Compound interest, annuity FV/PV, amortisation, VAT, income tax.

NEXT LEVEL LEARNING

If you have mastered this guide, you are fully prepared for the EUEE. This section gives you a glimpse of where mathematics goes at university — not to overwhelm you, but to show you how exciting the next chapter is.

Limits and Continuity

In Grade 12, you used derivative rules directly. At university you will prove those rules from first principles using the formal limit definition: $f'(x) = \lim_{h \rightarrow 0} [f(x+h) - f(x)] \div h$. You will also study when and why functions are continuous — the foundation of all calculus.

Multivariable Calculus

You differentiated $f(x)$ — a function of one variable. University calculus extends this to functions like $f(x, y) = x^2 + 3xy + y^2$. Partial derivatives measure the rate of change in one direction at a time. Used in economics, physics, and machine learning.

Differential Equations

A differential equation involves a derivative in the equation itself, for example $dy/dx = k \times y$ (exponential growth model). These equations model population growth, heat transfer, electrical circuits, and more.

Probability and Advanced Statistics

Grade 12 covers descriptive statistics. University adds probability theory, random variables, distributions (Normal, Binomial, Poisson), hypothesis testing, and regression — essential for research and data science.

Linear Algebra

You used basic matrices. University linear algebra covers vector spaces, eigenvalues, eigenvectors, and large systems of equations. It is the mathematical backbone of computer graphics and artificial intelligence.

Non-linear Optimisation

LP is just the beginning. At university you study the Simplex Method, non-linear programming, and convex optimisation — tools used by engineers, economists, and AI

researchers every day.

A Final Word

Mathematics is not about memorising formulas — it is about developing the habit of thinking clearly and systematically. Every formula in this guide comes from a logical argument. As you study, always ask *why* a formula works, not just *how* to apply it.

You have worked hard to reach Grade 12. The EUEE is your opportunity to show what you are capable of. Trust your preparation, stay calm, and give it your best. Good luck!